

Class 1.5 — Control Box I

Meeting date: Sunday, July 12, 2026 · 3 hours · Mr. Holcombe's house

T-MINUS STRIP

T-2 WEEKS

Confirm SeaMATE PufferFish Practice Board Kits are in hand (one per student). Re-stock soldering consumables (solder, tip-cleaner, heatshrink) if low.

T-1 WEEK

Confirm the 12V SLA battery + charger + leads are in hand (gating item for 1.6, not 1.5 — but the order has to be placed now). Charge the iPad.

T-1 DAY

Stage the workstations: one iron per kid, one PufferFish board per kid, the TriggerFish board in its tray with the Guide PDF loaded on the iPad. Test every iron.

T-0 (MEETING DAY)

Doors at 2:45 PM, brief at 3:00 sharp. Safety glasses laid out at every station. Fume extractor running, fan placed, multimeter and watt meter on the bench.

ORDER AHEAD · LEAD-TIME ALARMS

- SeaMATE PufferFish Practice Board Kits (one per student) — ~\$15 / kit × 12 = \$180 · by T-2 weeks
- **12V SLA battery + charger + leads (gating for 1.6 power-up) — ~\$80 all-in · by T-1 week**
- Pre-terminated tether (if not already received from Tier 3 order) — \$600 (already in BoM, confirm arrival) · by T-1 week
- Solder + heatshrink top-up if low — ~\$25 · by T-2 weeks

STAGE IN THE ROOM

- Soldering iron at each workstation (one per kid).
- One PufferFish practice board per kid, components staged next to it.
- TriggerFish REV 4 board in its tray; SeaMATE Guide queued on the iPad at the next block of components.
- Multimeter on every two stations; watt meter on the central bench.
- Safety glasses, helping hands, tip cleaner, flux pen, solder spool at every station.
- Fume extractor and fan positioned to pull/push fumes AWAY from the student (not toward).

SET UP

- Bench layout: solo stations in a U-shape so the advisor and Safety Briefer can see every iron.
- Power: surge protector per pair of stations; no daisy-chained extension cords.
- Ventilation: window open if weather permits, plus the desk fan and the fume extractor running before the irons heat.
- Lighting: bright direct light at every station — solder joints can't be inspected in dim light.
- First-aid kit on the bench, visible to the room.

CUE UP

- SeaMATE TriggerFish ROV Guide loaded on the iPad — bookmark the slide where 1.3 left off (post-fuse, post-caps).
- Slide deck 1.5 cued on the TV (this deck).
- Reference: "How to Solder Components to a PCB" (MATE presentation) — open in a browser tab for any kid who wants a refresher.
- Phone or camera ready for the Photographer.

STUDENTS BRING

- Closed-toe shoes.
- Hair tied back if long.
- A water bottle (they'll be at the bench for ~2.5 hours).
- Their notebook if they're keeping one.

CONFIRM · GATING

- 12V SLA battery + charger + leads in hand BEFORE 1.6 — order placed by T-1 week of 1.5 at the latest.
- Pre-terminated tether arrived; if not, escalate.
- Every kid wearing safety glasses at the iron — no exceptions.
- Fume extractor on before any iron heats.

PREP OWNERS

- Advisor — Order PufferFish kits + 12V SLA battery + charger · by T-2 weeks
- Advisor — Stage the workstations, test every iron · by T-1 day
- CEO (or designee) — Confirm Friday/Sunday attendance + workstation count match · by T-1 day
- Photographer — Charge phone/camera; confirm storage space · by T-0
- CFO (or designee) — Confirm 1.4 expenses logged in the BoM · by T-1 day